

MATH 4790: Quiz 3

February 20, 2026

Name: _____

Score: _____/15

Instructions: Show all work and justify your answers. Only calculators approved for Exam FM are permitted. Credit will not be given to just guessing the correct answer.

Question 1.

A borrower takes out a loan of \$5,620 to be repaid with 10 annual payments made at the end of each year where the first payment is P . The annual effective interest rate is 6%.

The first five payments increase by 3% each year and after the fifth payment, the payments decrease by 2% each year for the next five years.

Find the value of P .

(A) 700

(B) 720

(C) 740

(D) 780

(E) 800

Question 2.

At an annual effective interest rate of 5%, Fred and Carl have payment arrangements with the same present value.

- Fred receives a perpetuity due with annual payments of P .
- Carl receives an increasing annuity with annual end-of-year payments for 10 years. The payment at time t is

$$P + 65(t - 1), \quad t = 1, 2, \dots, 10.$$

Find P .

- (A) 120 (B) 155 (C) 185 (D) 210 (E) 265

Question 3.

A perpetuity pays \$100 at the end of each month for the first two years and then switches to annual end-of-year payments of \$1000. The annual interest rate during the first two years is 6%, and i thereafter. If the present value of the perpetuity is \$22,038, find i .

- (A) 3.5% (B) 3.75% (C) 4% (D) 4.25% (E) 4.5%